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## People use features, not programs

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**Problem: People are not computers. They use features rather than programs to characterize things.**

Indeed, people use **relevant features** to characterize events (rather than producing compact programs that generate those events from scratch). Suppose we are talking about a person born in the US in 2002.

- We want to mention that the person's given name is Shea. There were about 0.04% children with that name. If you regard causality in this case as a mere choice among options, **causal complexity** is about  $\log_2(1/0.0004) \approx 11$  bits.
- Now mentioning the name comes at a price on the **description complexity** side. 'Shea' appears 900<sup>th</sup> or so in the list of names ranked by popularity. With only this source of information, we have to pay  $\log_2(900) \approx 10$  bits of description complexity for the mention of the name.

As we can see, the mention of the name hardly contributes by itself to making that person unexpected.

- Suppose now that the name is 'Madonna' instead of 'Shea'. This would have made this person unique among 4 millions for that birth year. For an observer aware of name distributions, causal complexity amounts to  $\log_2(4 \times 10^6) = 22$  bits. On the other hand, the name 'Madonna' is at least as popular as 'Shea' on the Web, which makes the former no more complex than the latter. Mentioning the first name 'Madonna' for a person born in 2002 is an efficient way of making her quite **unexpected**, as it reduces complexity from 22 (causal) bits down to less than 10 (description) bits.

**Question:** Can features be relevant?

**Answer:** Yes, and this is precisely how people use language to talk about events.

$f \circ x(F)$  .

'Jeannine', L) .

**constraints** on

as predicates  $f_i(s)$  that  
true about situation

**nnlexity** of a feature

**Question:** Do unexpected features make events relevant?

**Answer:** Yes, thanks to the chain rule.

Let's observe how features' unexpectedness is transferred to the situation. Suppose we recount an event  $s$  by mentioning an **unexpected feature**  $f(s)$  (e.g. a fox wandering in the city). On the causality side, we have to measure the complexity needed by the "world" to make  $f(s)$  true. Contrary to mere description that only cares about  $f$ , causality requires to take both  $f$  and  $s$  into account to compute the complexity of reaching a state of the world in which  $f(s)$  holds.

- The chain rule for **causal complexity** reads:

*Chain rule for feature causality*

$$C_w(s) = C_w(f(s)) + C_w(s \mid f(s))$$

Note that  $f(s)$  expresses a constraint. It is not a ‘function’, but a logical predicate. This is why one may write  $C_w(s \mid f(s))$  even if  $s$  is not fully characterized at this stage of the computation. Equality comes from the fact that  $f$  is the only available means to compute the causality of  $s$  at this point.

$C_w(f(s))$  measures the complexity of generating a state of the world in which  $f$  is true when applied to the situation  $s$ .  
 $C_w(s \mid f(s))$  measures the causal complexity of other aspects of  $s$ .

- The chain rule for **description complexity** reads:

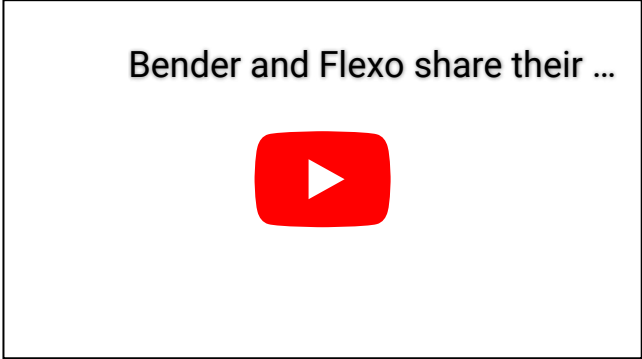
*Chain rule for feature description*

$$C_d(s) \leq C_d(f) + C_d(s \mid f(s))$$

Note that  $f$  appears alone in  $C_d(f)$ . This is a crucial point: description complexity is about the **conceptual** nature of features, whereas causal complexity is about making them true.

**Relevant features** are simple to describe, but hard to produce (as in our example about the fox). Features can be relevant for an observer while being **irrelevant** for another one. For instance, in this excerpt from the animated series Futurama, the two robots are amazed at a mathematical coincidence that leaves their friends completely unmoved.

[I found initially that the second serial number 2716057 is not the sum of two cubes and that it should read  $7216057 == 30^3 + 193^3$ ; however,  $2716057 == 952^3 + (-951)^3$  - Thank you [EJj](#) 😊]



- We can derive a chain rule for **unexpectedness**:

*Chain rule for unexpectedness*

$$U(s) \geq U(f(s)) + U(s \mid f(s))$$

where  $U(f(s)) = C_w(f(s)) - C_d(f)$ .

This chain rule, when repeatedly applied, offers an optimal guide for (artificial) storytelling: choose features and feature order that will eventually maximize unexpectedness.

## Features

0/1 point (ungraded)

Suppose we mention now (i.e. in the 2020s) the first name of a person born in the US in 2002. How unexpected is this feature if the first name is 'Michelle'?

- ☐ Not unexpected
- ☐ 1 bit  
✓
- ☒ 3 bits
- ☐ 7 bits

✗  
**Answer**  
Incorrect:  
The rank of "Michelle" among first names nowadays is about 200<sup>th</sup>, so its description can be assumed to cost about 8 bits. On the other hand, 0.25% girls were born with that name in 2002 (pay attention to the pink scale), which makes causal complexity close to  $\log_2(1/0.0025) \approx 9$  bits. So the complexity drop is about 1 bit.

 Answers are displayed within the problem

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